

HEMANATHAN G *EMBEDDED ENGINEER*

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OBJECTIVE

Technical Embedded Systems Engineer with strong expertise in hardware design, firmware development, and sensor-to-cloud IoT integration. Demonstrated success in leading commercial hardware projects from concept to production, managing PCB layout, assembly, and system validation.

SKILLS

Programming: C, Embedded C
Microcontrollers: ESP32, ESP8266, STM32 (Fundamentals), Arduino
Communication Protocols: UART, SPI, I2C, CAN
PCB Design & EDA Tools: KiCad, OrCAD, Gerber Generation, BOM Management
Hardware & Power Engineering: Power Analysis, Schematic Capture, Component Selection, Hardware Debugging & Testing
IDEs & Tools: VS Code, Arduino IDE, STM32CubeIDE.
Core Embedded Concepts: GPIO, Timers, Interrupts, PWM, ADC, DAC, Sensor Interfacing

CERTIFICATIONS

VAC in Embedded Systems & IoT: Taras System And Solution | Aug 2025

VAC in Industrial Advanced PCB Design: Wizard Systems | Jun 2025

AREA OF INTERESTS

- Embedded Hardware Engineering
- PCB Design
- Microcontroller Firmware Development
- B2B product Prototying

EDUCATION

Kongunadu College of Engineering and Technology, Thottiam, Trichy, *B.E ECE*
2023 – 2027 | Thottiyam , Trichy
CGPA: 8.27

Raj Vidya Bhavan Matric. Hr. Sec. School, *HSC*
2022 – 2023 | Thuraiyur , Trichy
Percent:87%

Raj Vidya Bhavan Matric. Hr. Sec. School, *SSLC*
2020 – 2021 | Thuraiyur , Trichy

EXPERIENCE

EMBEDDED SYSTEM INTERN, *WATTSTRONS*
08/2026 | BENGALURU,KARNATAKA
End-to-End B2B Hardware Development: Executed full-lifecycle hardware design for 2 B2B product lines from concept to production, including component selection , power analysis, datasheet reference analysis, schematic creation, and BOM management.
PCB Design & Fabrication Workflow: Managed complete PCB development cycle including component placement, trace layout, Gerber file generation, fabrication setup, component assembly, and multi-stage layout/schematic reviews.

PROJECT

Smart Parking System:
- Engineered an automated IoT parking management system integrating ESP32 sensor arrays, computer vision algorithms for license plate and spot detection, and a live web telemetry dashboard for real-time occupancy updates.

Intelligent Vision & Remote Monitoring System for Food Prediction:
- Designed an IoT-driven food degradation monitoring platform combining environmental sensor arrays (gas, temperature, humidity) with image processing models to analyze spoilage trends and deliver web-based predictive alerts.

Autonomous Horticulture Robot for Plant Monitoring:
- Built an autonomous agricultural mobile robot featuring multi-sensor acquisition (soil moisture, pH, temperature), automated feedback pump control for precision watering, and cloud-based monitoring.

ACHIEVEMENTS

1st Prize – TEZARIO'26 (Dept. of IT, KNCET): To secure 1st place by engineering an IoT-based Foot Pressure Monitoring & Orthotics Adjustment system.
2nd Prize – SCHWAN TECH 3.0 (Dept. of BME, KNCET): To win 2nd place for developing a Wearable Bio-Signal Monitoring System.
3rd Prize – GENESIS'26 (Dept. of CSE, KNCET): Won 3rd place for designing an automated Smart Parking Management System.
Best Innovation Award – Sri Eshwar Thiraan 2025: To win the Special Prize for Best Innovation with an Autonomous Horticulture Robot.